

Loan Approval Prediction Report

1. Objective

The goal of this project is to build a supervised machine learning model to predict loan approval using borrower features. The focus is on preprocessing, handling class imbalance, evaluating multiple models, and providing business-oriented interpretation of results.

2. Data Preprocessing

- **Missing values:** Numerical features filled with median; categorical features filled with mode.
- **Encoding:** Label encoding applied to categorical variables.
- **Scaling:** StandardScaler used for numerical features.
- **Train/Test split:** 80/20 with stratification to preserve class distribution.

3. Class Imbalance

- Target variable (Loan_Status) showed imbalance (~70% approved vs ~30% not approved).
- **SMOTE** applied to oversample the minority class, resulting in balanced training data.
- Alternative approaches (class weights, undersampling) were considered.

4. Models Compared

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost (optional advanced model)

5. Visualizations

- Confusion matrix heatmap (errors distribution).
- ROC curve (overall model quality).
- Precision-Recall curve (threshold tuning).
- Feature importance chart (borrower attributes driving predictions).

6. Business Interpretation

- **Precision vs Recall trade-off:**

- High recall ensures fewer eligible borrowers are wrongly rejected.
 - High precision reduces risk of approving unqualified borrowers.
- **Threshold tuning:** Recommended threshold ~0.4–0.5 to balance recall and precision.
- **Feature importance:** Income, loan amount, and credit history are the strongest predictors.

7. Recommendation

Deploy the Random Forest model with a tuned threshold around 0.45. This maximizes recall (catching more eligible borrowers) while keeping precision acceptable for business risk management. Logistic Regression can serve as a baseline due to interpretability.

Short Report: Model Trade-offs and Suggested Threshold

Model Performance Overview

The following table summarizes the performance of the trained models based on key evaluation metrics:

Model	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8370	0.9059	0.8701	0.8220
Decision Tree	0.8169	0.6824	0.7436	0.6701
Random Forest	0.8556	0.9059	0.8800	0.7565

Model Trade-offs

- **Random Forest** emerged as the top performer with the highest F1-Score (0.8800) and Precision (0.8556), and excellent Recall (0.9059). This model is highly effective at identifying positive cases while keeping false positives low, making it a strong choice for balancing risk and reward in loan approvals.
- **Logistic Regression** offered a competitive performance with a high ROC-AUC (0.8220) and strong Recall, coupled with interpretability which is often crucial in financial contexts.

- **Decision Tree** showed lower performance across most metrics, indicating it might not be the most suitable model for this problem without significant optimization.

Suggested Deployment Threshold

Based on the analysis, a classification probability threshold of **0.5** is suggested for initial deployment using the **Random Forest** model. This threshold provides a robust balance between maximizing the approval of eligible loans (high recall) and minimizing the approval of defaulting loans (good precision).

For further refinement, a **detailed cost-benefit analysis** (as demonstrated in the previous section) is essential. This analysis would involve quantifying the monetary costs and benefits associated with true positives, true negatives, false positives, and false negatives. By simulating outcomes at various thresholds, the optimal threshold can be identified to align precisely with specific business objectives and risk tolerance, maximizing net profit or minimizing overall loss for the institution.
